STAT 525 — Spring 2009

Midterm 1

Thursday February 19, 2009

Time: 2 hours

Name (please print): ____

Show all your work and calculations. Partial credit will be given for work that is partially correct. Points will be deducted for false statements, even if the final answer is correct. Please circle your final answer where appropriate.

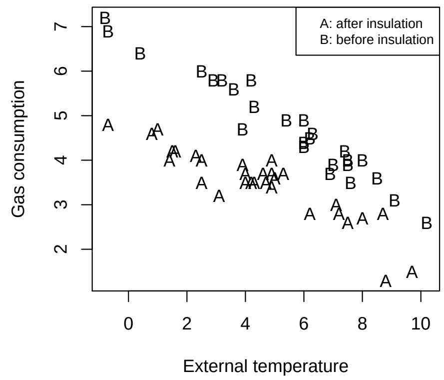
This exam is closed-book. You may consult one page with your hand-written notes. Calculators are permitted.

Honor code: I promise not to cheat on this exam. I will neither give nor receive any unauthorized assistance. I will not to share information about the exam with anyone who may be taking it at a different time. I have not been told anything about the exam by someone who has taken it earlier.

Signature: ____ Date: _____________

1. Mr Derek Whiteside of the UK Building Research Station recorded the weekly gas consumption (in 1000s of cubic feet) and average external temperature (in Fahrenheit) at his house in south-East England for 56 consecutive weeks, before and after a house insulation was installed.

   The figure below shows the plot of the observed data. The goal of the study is to assess the effect of the insulation on gas consumption.



   (a) **(4 pts)** Is this an observational study or a designed experiment? Would you recommend that a regression or a correlation model be used?

   Answer:

   This is an observational study, and values of `temperature` *cannot be fixed in advance. Therefore a correlation model is appropriate. However the model is equivalent to a simple linear regression where inference on* `insulation` *is made conditional on the observed values of* `temperature`*.*

   (b) **(8 pts)** The researcher considers to model gas `consumption` as a first order linear regression with interaction, using both `temperature` and `insulation` as predictors. For `insulation`, he created a categorical variable that takes the value of 1 for "after installing the installation", and 0 otherwise.

   State the linear model and *all the assumptions*, and interpret the parameters.

   Answer: *Linear model:*

$$Consumption_i = \beta_0 + \beta_1 temperature_i + \beta_2 insulation_i + \beta_3 temperature_i \cdot insulation_i + \epsilon_i$$

   where:

   - ϵ_i*: independent* $N(0, \sigma^2)$ *random variables.*
   - *Temperature and insulation are considered fixed for each consumption observation.*
   - *There is a linear relationship between the expected value of consumption, temperature and insulation.*

   Parameter interpretation:

   - β_0*: intercept when* $insulation = 0$
   - $\beta_0 + \beta_2$*: intercept when* $insulation = 1$
   - β_1*: change in the consumption mean when the temperature increases 1 unit and* $insulation = 0$.
   - $\beta_1 + \beta_3$*: change in the consumption mean when the temperature increases 1 unit and* $insulation = 1$.

(c) **(8 pts)** Below is a partial SAS output of the linear model in (b).

Source	DF	Sum of Squares	Mean Square
Model	?	?	?
Error	52	5.42525	0.10433
Corrected Total	55	75.01429	

i. Complete the ANOVA table by replacing the "?" marks.
Answer:

$$DF = 55 - 52 = 3$$
$$SSR = 75.01429 - 5.42525 = 69.58904$$
$$MSR = \frac{69.58904}{3} = 23.19635$$

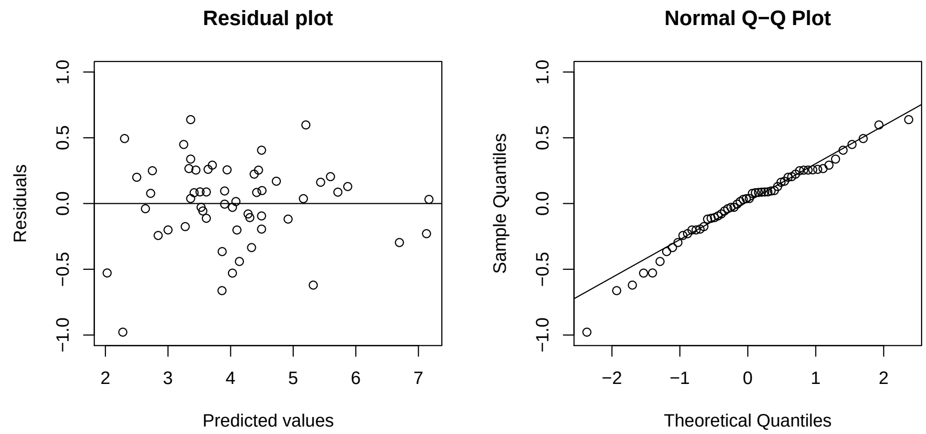
ii. Conduct the overall test of whether there is a regression relation between the response and the predictors. State the null hypothesis and the alternative, conduct the test at the significance level of 95%, and interpret the results.
Answer: *H_0: All β_i=0. H_a: at least one $\beta_i \neq 0$. The corresponding F statistic is $F = 222.33$. Since $F > F(3, 52) = 2.7826$, then we have enough information to reject H_0. (there is a regression relation between the response and some of the predictors)*

iii. Calculate and interpret the coefficient of multiple determination.
Answer:

$$R^2 = \frac{SSR}{SSTO} = \frac{69.58904}{75.01429} = 0.9277$$

The linear model is explaining a 92.77% of the total variation of the data.

(d) **(8 pts)** The figures below show results of residual analysis of this model.



i. Interpret the residual plot, and make a conclusion regarding the model fit.
Answer: *The residual plot does not have any evident pattern and the assumptions of independence and constant variance can be assumed here. It seems that the model is doing a reasonable fit on the data*

ii. What are the quantities shown on the x and y axes of the quantile-quantile plot? What is the conclusion from the plot for this dataset?
Answer:

- *X axis: quantiles of the expected residuals under the normality assumption.*
- *Y axis: empirical quantiles of the residuals.*

The QQ-plot shows that we can assume that the residuals are normally distributed. They do not present a large departure from normality.

iii. Explain why one someone might question the independence assumption in regards to this analysis.

Answer:

Experimental measurements are taken during 56 consecutive weeks. Therefore the assumption of independence of the measurements may not hold, and the errors may have a correlation structure.

(e) **(8 pts)** To formally check the quality of the model fit, the researcher conducts a lack of fit test. He combines the observations into 10 groups of roughly equal size; observations in each group have the same `insulation` status, and similar values of `temperature`. He then fits a model using the group as a categorical predictor. The SAS output of this model is given below.

Source	DF	Squares	Mean Square	F Value
Model	9	70.10396429	7.45599603	43.36
Error	46	4.91032143	0.17196351	
Corrected Total	55	75.01428571		

State the null and the alternative hypotheses for the lack of fit test. Conduct the test at the significance level of 95%, and interpret the result.

Answer: *Alternatives:* $H_0 : E[Y] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_2 \cdot X_3$ *vs* $H_a : E[Y] \neq \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_2 \cdot X_3$.

The test statistic is:

$$F^* = \frac{\frac{SSE_R - SSE_F}{df_R - df_f}}{\frac{SSE_F}{df_F}} = \frac{\frac{5.42525 - 4.91032143}{6}}{\frac{4.91032143}{46}} = 0.803977$$

And we reject H_0 *if* $F^* > F(.95, 6, 46) = 2.303509$.

Conclusion: we fail to reject H_0*, therefore there is no evidence against the proposed model.*

(f) **(8 pts)** The output below provides parameter estimates of the model in (b), and the associated sums of squares. `insutemp` in the output corresponds to the `insulation x temperature` interaction term.

Test whether the installed insulation has an effect on gas consumption. State the null and the alternative hypotheses, conduct the test at the significance level of 95%, and interpret the result.

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	Type I SS	Type II SS
Intercept	1	6.85383	0.13596	.	.
temp	1	-0.39324	0.02249	35.01942	31.90543
insu	1	-2.12998	0.18009	33.22449	14.59416
insutemp	1	0.11530	0.03211	1.34514	1.34514

Answer: *Alternatives:* $H_0 : \beta_2 = \beta_3 = 0$, $H_a : \beta_2 \neq 0$ *or* $\beta_3 \neq 0$. *The test statistic is:*

$$F^* = \frac{\frac{SSR(X_2,X_3|X_1)}{2}}{\frac{SSE(X_1,X_2,X_3)}{52}} = \frac{\frac{SSR(X_3|X_1,X_2)+SSR(X_2|X_1)}{2}}{\frac{SSE(X_1,X_2,X_3)}{52}}$$
$$= \frac{\frac{33.22449+1.34514}{2}}{\frac{5.42525}{52}} = 165.6717$$

And we reject H_0 *if* $F^* > F(.95, 2, 52) = 3.1751$. *Then the installed insulation has an effect on gas consumption.*

(g) **(8 pts)** Next, the researcher would like to test whether a decrease in the `temperature` results in a weaker increase in `consumption` in presence of insulation. State the null and the alternative hypotheses, conduct the test at the significance level of 95%, and interpret the results (*Hint:* use a one-sided test; the null hypothesis represents the "status quo", and the alternative represents an "improvement").

Answer:

The null hypothesis of interest is "status quo", i.e. the insulation results in a same or stronger change in gas consumption when the temperature drops. The alternative hypothesis is the "improvement", i.e. a weaker change in gas consumption when the temperature drops.

This is translated into $H_0 : \beta_{\texttt{insutemp}} \leq 0$ *vs the alternative* $H_a : \beta_{\texttt{insutemp}} > 0$.

The test statistic $t_s = \frac{0.11530}{0.03211} = 3.59 > t_{52}^{1-0.05} = 1.67$, *therefore we reject* H_0.

(h) **(8 pts)**

i. Give the interpretation of a 95% confidence interval associated with the parameter `insulation x temperature` (you do not need to calculate the interval).

Answer: *Under repeated sampling, a 95% of the corresponding confidence intervals will contain the true value of* β_3 *(increase in the rate of change of consumption in terms of temperature under the presence of insulation)*

ii. Can the confidence interval be used to answer the same question as in (g)? Will the conclusion always agree with the conclusion in (g)?

Answer:

No. The interpretation a confidence interval always supports the conclusion of a two-sided test. However, we have a one-sided test in part (g). Therefore it is possible that a confidence interval includes 0, however we reject $H_0 : \beta_{12} = 0$ *with a one-sided test.*

(i) **(8 pts)** Describe the procedure that you'd use to calculate the power of the test in (g) when the true difference in slopes is 0.2. Use $\alpha = 0.05$, and the estimate of standard error from this dataset. Provide the exact formula for the power (you do not need to give a numeric answer).

Answer:

$Power = P\{t_s > t_{56-4}^{1-0.05} \mid \frac{0.2}{0.03211}\}$, *where* $t_s \sim Student_{56-4}$.

(j) **(8 pts)** Produce the predicted mean of `consumption`, and the associates standard error, for `temperature` of 10 Farenheit in absence of insulation. Use the parameter estimates in (f), and the variance-covariance matrix below.

Covariance of Estimates

Variable	Intercept	temp	insu	insutemp
Intercept	0.018486202	-0.002705317	-0.018486202	0.0027053168
temp	-0.002705317	0.0005056667	0.0027053168	-0.000505667
insu	-0.018486202	0.0027053168	0.0324330268	-0.005050896
insutemp	0.0027053168	-0.000505667	-0.005050896	0.0010311886

Answer: *Take*

$$X_h = \begin{bmatrix} 1 \\ 10 \\ 0 \\ 0 \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 6.8538 \\ -0.3932 \\ -2.12998 \\ 0.11530 \end{bmatrix}$$

then

$$\hat{Y}_h = X_h' \mathbf{b} = 2.9218$$
$$\hat{Var}(\hat{Y}_h) = X_h' \hat{Var}(\mathbf{b}) X_h = 0.014947.$$

(k) **(8 pts)** The predicted mean of `consumption`, and the associates standard error, are given below for two more configurations of the independent variables.

temp	insul	predicted	st. error
6.0	0	4.4944	0.0650
8.0	0	3.7079	0.0870

i. Produce the simultaneous 95% confidence intervals for the population means of these two configurations. (*Hint:* use the procedure with the narrowest confidence intervals).

Answer:
The Bonferroni multiplier is $t_{52}^{1-0.05/(2\cdot 2)} = 2.308$.
The Working-Hotelling multiplier is $\sqrt{4F(1-\alpha, 4, 52)} = 3.1935$.
Since the Bonferroni multiplier is the smallest, we use it to construct the confidence intervals.
The actual intervals are $(4.4944 \pm 2.308 \cdot 0.0650) = (4.34438, 4.64442)$
and $(3.7079 \pm 2.308 \cdot 0.0870) = (3.507104, 3.908696)$

ii. What multivariate random event is controlled by the simultaneous confidence intervals above?

Answer: *Using the Bonferroni procedure we are providing a lower bound to the confidence coefficient for the two configurations simultaneously. Then we are controlling the probability that both confidence intervals will contain their population means at the same time.*

2. Short answer questions. Each part is unrelated.

(a) **(8 pts)** A researcher is designing a new experiment involving an independent variable X and a responce Y. Data from the experiment will be analyzed using a linear regression. He considers two options for the choice of values of X:

`1) X = 1 2 3 5 5 7 8 9` and `2) X = 1 1 1 5 5 9 9 9`

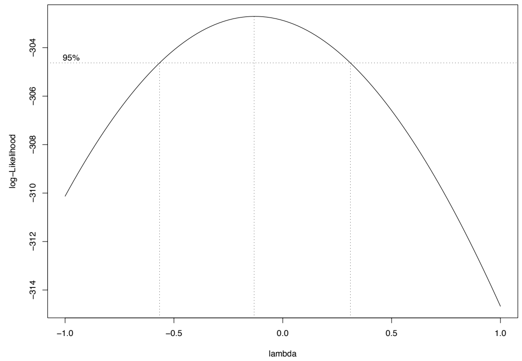
Which design would you recommend and why? If there is a situation favorable to each, please specify.

Answer:

In a simple linear regression, $var\{b_0\}$ *and* $var\{b_1\}$ *are inversely proportional to* $\sum_{i=1}^{n}(X_i - \bar{X})^2$. *Therefore, if the linear model is appropriate,* X_i *taken far from* $\bar{X}$ *will result in more accurate parameter estimates. Therefore design 2) is preferred as compared to 1).*

At the same time, if the nature of the relationship between X *and* Y *is unknown, it is important to take measurements at intermediate values of* X *as this will allow us to estimate more flexible functional forms. In this case design 1) is preferred.*

(b) **(8 pts)** The figure below shows a Box-Cox plot from a simple linear regression. Explain how to use this plot, and what, if any, actions should be taken for the data in this analysis.



Answer:

The Box-Cox procedure fits the model $Y_i^\lambda = \beta_0 + \beta_1 X_i + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$, *and simultaneously estimates* β_0, β_1, σ^2 *and* λ *using Maximum Likelihood. The estimated value of* λ *is used to determine the appropriate transformation of* Y.

Here $\hat{\lambda}_{ML} \approx -0.15$. *However* $\hat{\lambda}_{ML} = 0$ *is within the range of variability of* $\hat{\lambda}_{ML}$, *and therefore* $\hat{\lambda}_{ML} = 0$ *is an acceptable transformation. By definition,* $Y^\lambda = log_e Y$ *when* $\lambda = 0$. *Therefore a log transformation of* Y *can be used.*